COURSE TITLE	**LINUX ADMINISTRATION**
COURSE CODE	**01CT2509**
COURSE CREDITS	**4**

Objective:

1 To impart knowledge and skills on various practical and theoretical aspects of Linux operating system (OS) basics and Linux OS based server configuration, management and administration.

Course Outcomes: After completion of this course, student will be able to:

1 Understand Linux utilities to create and manage simple file processing operations and Linux boot processing (Understand)

2 Apply Command line in Linux to manage user, user groups, system management, volume management, and troubleshooting application, scheduling task and system level issue (Apply).

3 Illustrate client server applications with appropriate security (Apply).

4 Configure various services of Linux such like DNS, Apache web server, virtualization (Apply).

5 Evaluate various shell Scripting (Analyze).

Pre-requisite of course: Basics of Operating System

Teaching and Examination Scheme

Theory Hours	Tutorial Hours	Practical Hours	ESE	IA	CSE	Viva	Term Work
3	0	2	50	30	20	25	25

Contents : Unit	Topics	Contact Hours
1	**Introduction And Installation** Linux introduction and file system, Basic Features, Advantages, Installing requirement, Basic Architecture of Unix/Linux system, Kernel, Shell., Linux File System - Boot block, How Linux access files, storage files, Linux standard directories, Download, install, update, Manage software packages from Red Hat, Yum package repositories..	5

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Contents : Unit	Topics	Contact Hours
2	**Management of Files Using Command Lines** Introduction to BASH, Command-line shortcuts, File Types, Ownership and Permissions, File management and manipulation, Moving users & its directories, Miscellaneous Tools, Editors, Create and Edit text files with vim (open, edit, and save text files), Commands for files and directories cd, ls, cp, md, rm, mkdir, rmdir, pwd, file, more, less, creating and viewing files using cat, file comparisons – cmp & comm, View files, disk related commands, checking disk free spaces, regular expressions with grep	5
3	**Managing Users and Groups** Creating and managing user/s and group commands, User management tools, Users and Access Permissions, Updating users and group attributes	4
4	**Booting and shutting down** Boot Loaders, the init process, Rc scripts, enabling, Disabling services, Booting in recovery mode	4
5	**File Systems** Makeup of file systems, managing file systems, Adding a new disk, Volume Management, Creating file systems	3
6	**Core System Services** The init Daemon, xinetd and inetd, The Logging Daemon, Configuring Logging Daemon, The CRON program	5
7	**Compiling the Linux Kernel** Kernel concepts, Finding Kernel Source Code, Building the Kernel, Patching the Kernel	4
8	**DNS** Installing DNS Server, Configuring DNS server, DNS records types, setting up BIND database file, The DNS Toolbox, Configuring DNS clients	4
9	**Apache Web Server** HTTP Protocol, Installing Apache HTTP Server, Starting up and shutting down Apache, Testing Apache Installation, Configuring Apache, Troubleshooting Apache	4
10	**Virtualization** Virtualization Implementation, Kernel based Virtual Machines (KVM), 06 Containers, Docker, Kubernetes	4
Total Hours		**42**

Suggested List of Experiments:

Contents : Unit	Topics	Contact Hours
1	**Experiment 1** Linux introduction & Installation of Linux-Operating system.	2
2	**Experiment 2** Software package management, mount and unmount the disks, devices	2

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Suggested List of Experiments:

Contents : Unit	Topics	Contact Hours
3	**Experiment 3** File Handling Commands	2
4	**Experiment 4** User Handling Commands	2
5	**Experiment 5** Group Handling Commands	2
6	**Experiment 6** Startup and Shutdown Commands	2
7	**Experiment 7** Installation and Configuration of DNS server	2
8	**Experiment 8** Installation and Configuration of Apache Server	2
9	**Experiment 9** Building and patching Linux Kernel.	2
10	**Experiment 10** Write a command with syntax & usage and execute the advance filters such as grep, egrep, fgrep.	2
11	**Experiment 11** Write a command with syntax & usage then execute the ps command, process management commands: & nohub, kill, nice.	2
12	**Experiment 12** Write a command with syntax & usage then execute the communication commands.	2
13	**Experiment 13** To execute device pattern using Meta character to match each of the following.	2
14	**Experiment 14** File Moment Using Command Line Arguments	2
15	**Experiment 15** Configuring different servers (FTP, SSH, NFS, NTP Time Server, DHCP, Samba)	2
Total Hours		**30**

Textbook :

1 Linux Administration: A Begineer’s Guide, Steve Shah, Tata McGraw-Hill Publishing Company Limited, 2020

References:

1 REDHAT System Administration-1 Student Work book, REDHAT System Administration-1 Student Work book, Susan Lauber,, REDHAT Inc, 2014

Suggested Theory Distribution:

The suggested theory distribution as per Bloom's taxonomy is as follows. This distribution serves as guidelines for teachers and students to achieve effective teaching-learning process

Distribution of Theory for course delivery					
Remember / Knowledge	**Understand**	**Apply**	**Analyze**	**Evaluate**	**Higher order Thinking / Creative**
15.00	15.00	40.00	10.00	10.00	10.00

Instructional Method:

1 The internal evaluation will be done on the basis of continuous evaluation of students in the laboratory and class-room.

2 Practical examination will be conducted at the end of the semester for evaluation of performance of students in laboratory.

3 Students may use supplementary resources such as online videos, NPTEL videos, e-courses, Virtual Laboratory, etc.

4 The course delivery method will depend upon the requirement of content and need of the students. The teacher in addition to conventional teaching method (Chalk and Talk) may use any of the tools such as demonstration, role play, Quiz, brainstorming, Flipped class, Project based learning, Collaborative learning, MOOCs etc. for effective teaching.

Supplementary Resources:

1 https://www.tutorialspoint.com/linux_admin/index.htm

2 https://linode.com/docs/tools-reference/linux-system-administration- basics/ opensourceforu.com/2016/07/introduction-linux-system-administration/https://www.linuxfoundation.org

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